A RESEARCH PROJECT ON THE ORIGINS OF DIAMONDS

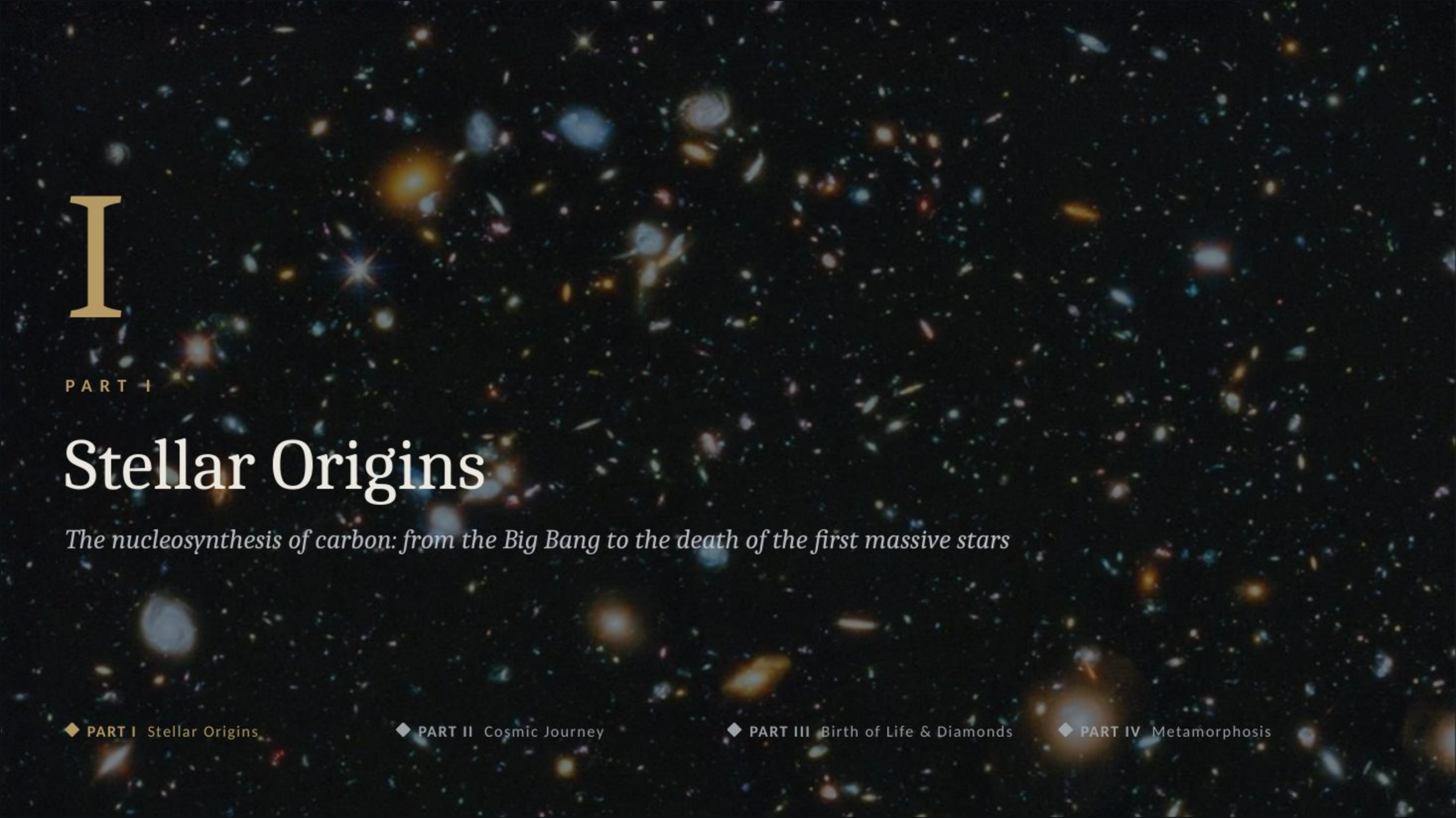
Birth of a Diamond

From stellar carbon to crystal, in four parts

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The research in one frame: following a single carbon atom across 13.8 billion years





I

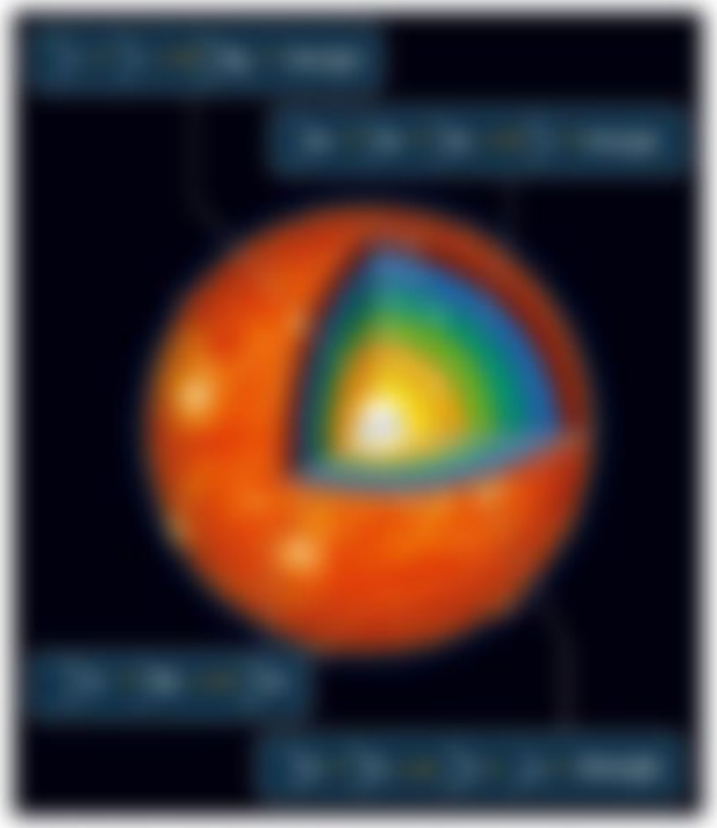
PART I

Stellar Origins

The nucleosynthesis of carbon: from the Big Bang to the death of the first massive stars

◆ **PART I** Stellar Origins ◆ **PART II** Cosmic Journey ◆ **PART III** Birth of Life & Diamonds ◆ **PART IV** Metamorphosis

Carbon is forged in stellar cores by the triple-alpha process



$3\,{}^{4}\mathrm{He} \longrightarrow {}^{12}\mathrm{C}$

- Hydrogen fusion (H → He) powers the main sequence. When the core exhausts hydrogen it contracts and heats until helium ignites.
- Helium burning happens in stars of roughly one solar mass and above, but only stars of ≳ 0.5 $M_\odot$ synthesize through carbon, oxygen and silicon burning in an onion core, and end in a core-collapse supernova.
- In red supergiants carbon is produced in large quantities and partly found ensnared into oxygen-, neon- and heavier elements as the star evolves.

10^8 K

core temperature

> 8 $M_\odot$

mass for full burning

^{12}C

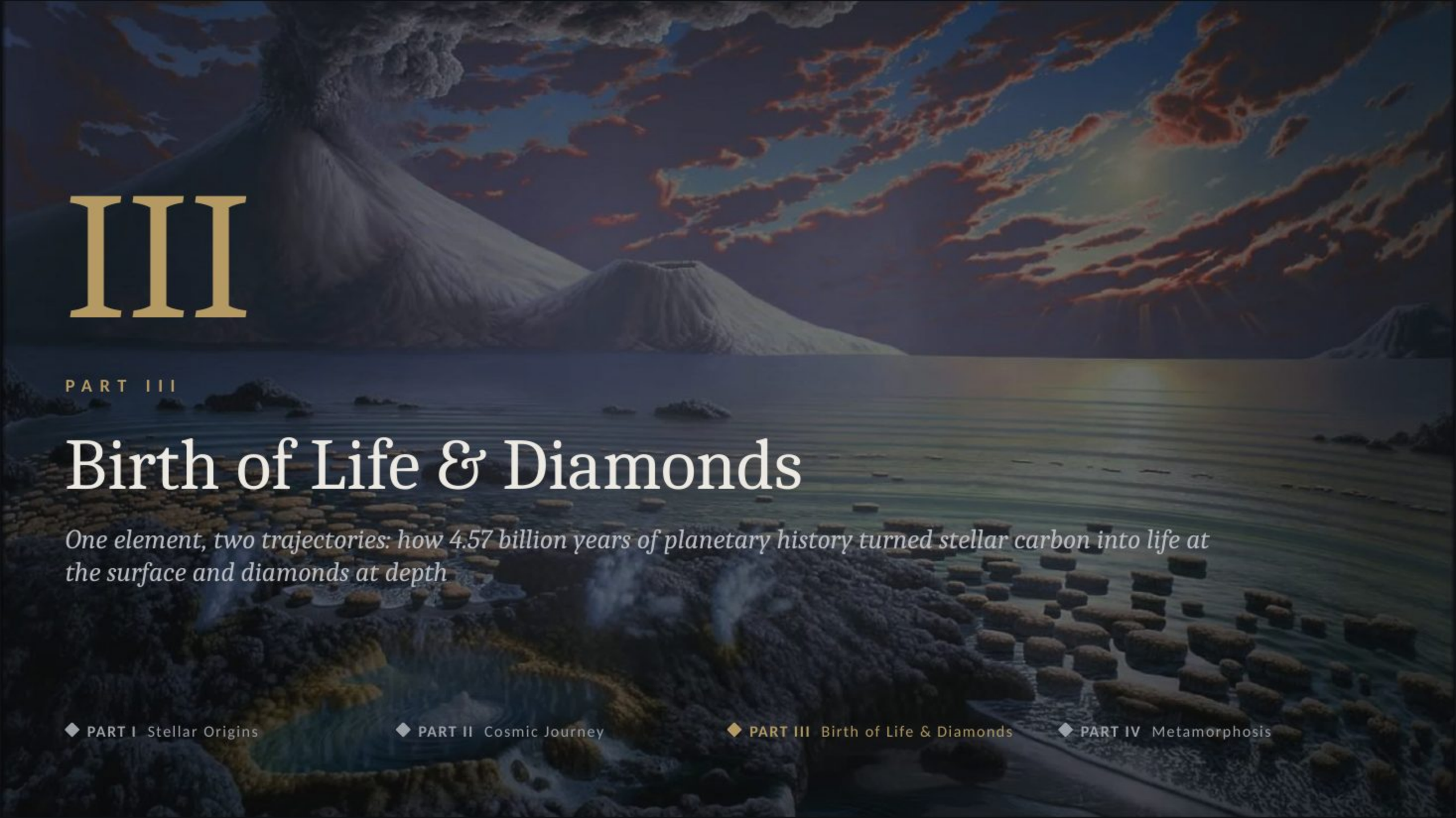
first stable carbon

II

PART II

Cosmic Journey

From stellar interiors to the protoplanetary disk: carbon spends 10^5 to 10^8 years in the interstellar medium



III

PART III

Birth of Life & Diamonds

One element, two trajectories: how 4.57 billion years of planetary history turned stellar carbon into life at the surface and diamonds at depth

Geological time at a glance: 4.57 billion years in one spiral



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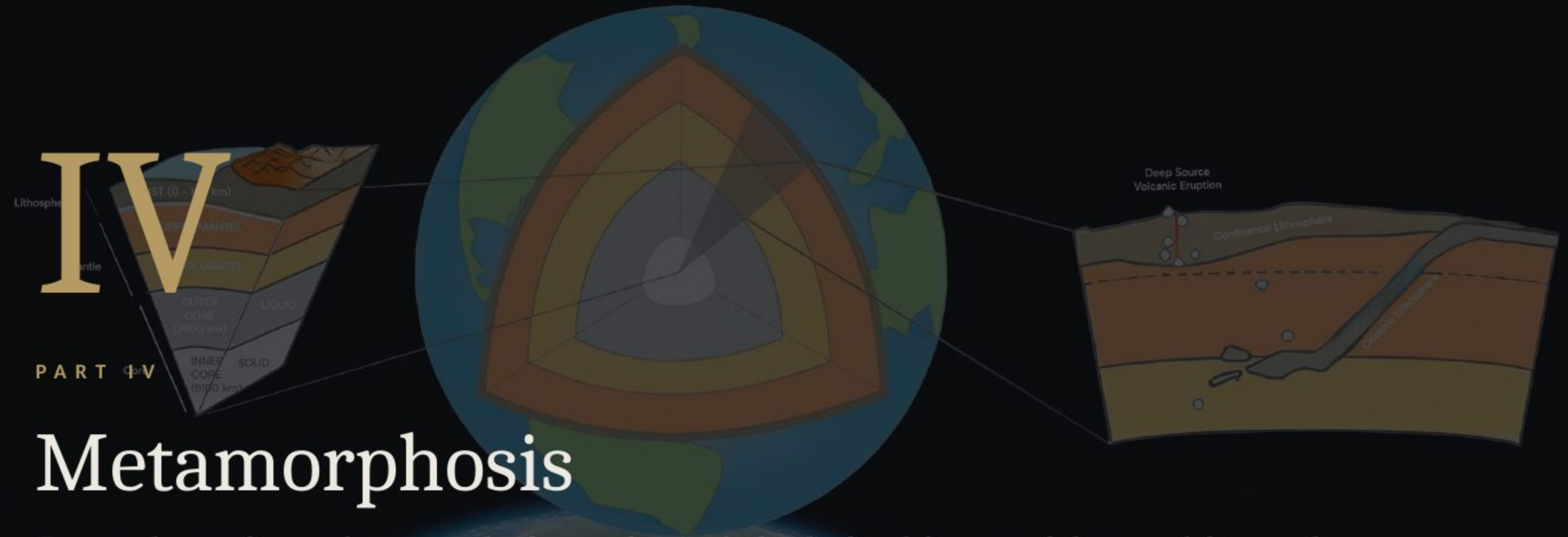
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IV

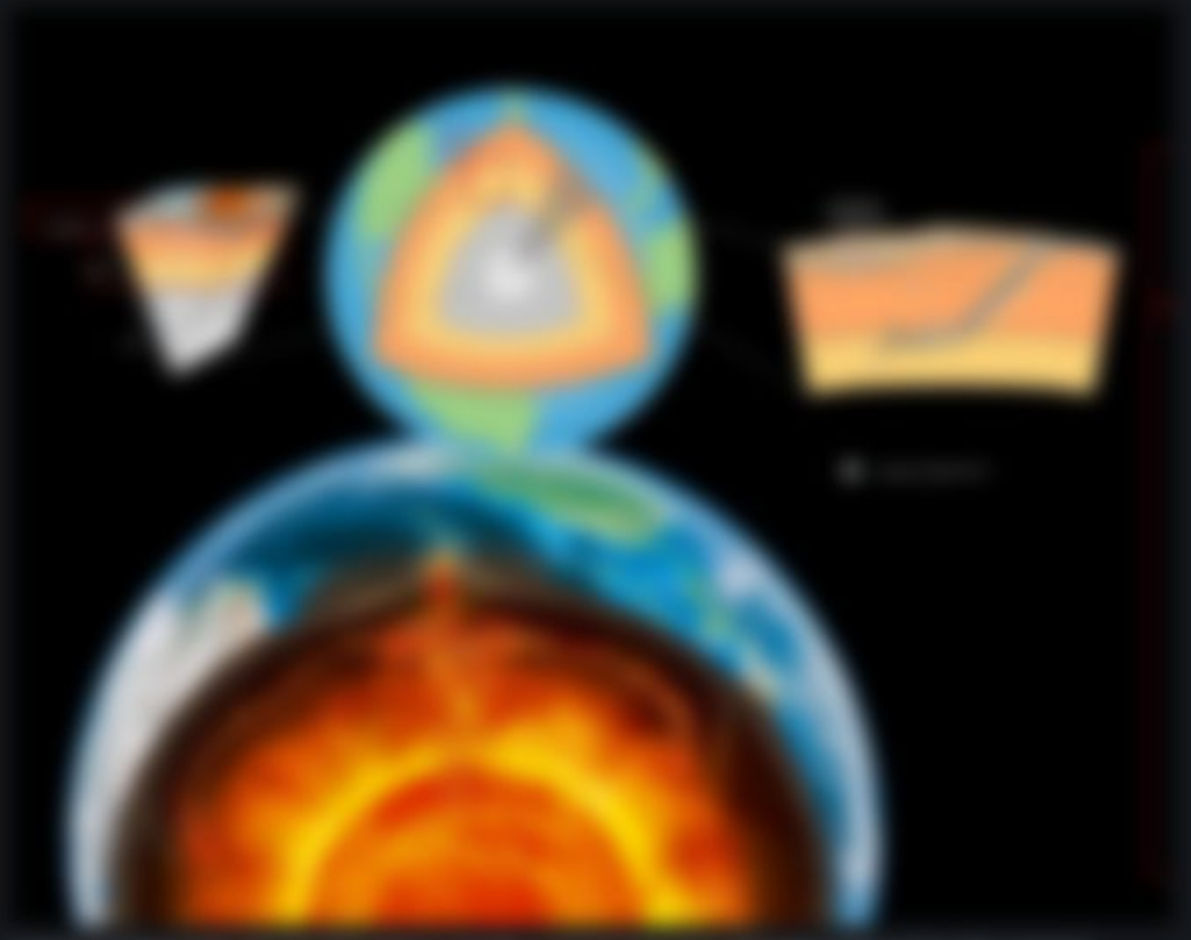
PART IV

Metamorphosis

From carbon to diamond: cratons, the deep carbon cycle, kimberlite delivery, and the growth history that makes each crystal unique

◆ PART I Stellar Origins ◆ PART II Cosmic Journey ◆ PART III Birth of Life & Diamonds ◆ PART IV Metamorphosis

Where diamonds form: a depth profile of the Earth





Kimberlites: Earth’s diamond delivery system, in seven steps



Kimberlite eruptions are the only known natural process that can bring diamonds from the deep mantle to the surface intact. Most diamonds are destroyed before they get there.

1. **Carbon in the mantle.** Around 150 km down, carbon dissolves into olivine-rich rock under extreme heat and pressure.
2. **Crystallization at depth.** At 150–250 km, carbon crystallizes directly as diamond in the lithospheric mantle.
3. **Long-term stability.** There, and in cratonic roots around the world, for 1–3 billion years.
4. **Kimberlite melt forms.** Deep mantle processes generate rare kimberlite melts that can rise through the lithosphere.
5. **Rapid upward transport.** Melts rise at tens of km per hour, 40 to 60 times faster than most magmas, and are powered by dissolved gas.
6. **Explosive emplacement.** Gas expands as the melt nears the surface, driving explosive eruptions, shattering wall rock and forming diatreme pipes.
7. **Exposure and mining.** Erosion exposes the pipes. Diamonds are mined directly from kimberlite deposits.

Diamonds are not formed during eruptions. The eruption is the last step in a journey that began billions of years earlier.

Each natural diamond is unique: a morphological catalogue (Williams, 1902)



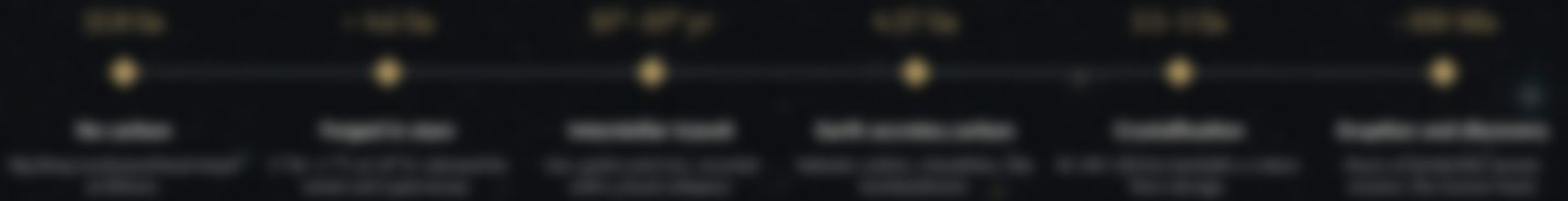
Rough diamond from the Kimberley mines, Transvaal, and C. J. Williams, The Diamond Mines of South Africa (1902), as cited in Gardiner's Adventure Corridor.

- **Crystal habit**
  Most rough is octahedral, cubic, or a mixture of the two, and the habit is the first clue to a stone's history.
- **Twinning**
  Contact twins such as the Macle group record a moment when two crystals grew together and then kept growing.
- **Colour**
  Colour zones often mirror the habit, tracing the shape of the crystal as it grew layer by layer.
- **Polycrystalline growth**
  Boart, ballas and carbonado are aggregates of many tiny crystals, with no cleavage and no transparency.
- **Irregular forms**
  Resorption, growth interruption, and fracture can leave a stone with no regular shape at all.

A diamond is not manufactured by people; each hundredth of a million of years, in an environment of extraordinary heterogeneity, the crystal is the cumulative, irreversible record of contingencies that will never converge the same way again.

◆ CONCLUSION

The biography of a diamond, in six moments



A diamond is older than the eruption that delivered it by one to three billion years, and its carbon is older than the star. The crystal in the band is the cumulative record of a single element's journey through stars, interstellar space, a churning planet and the deep mantle. No same journey that, at Hargreaves, produced the band that holds it.